A hierarchical framework for generalizations of measurement incompatibility

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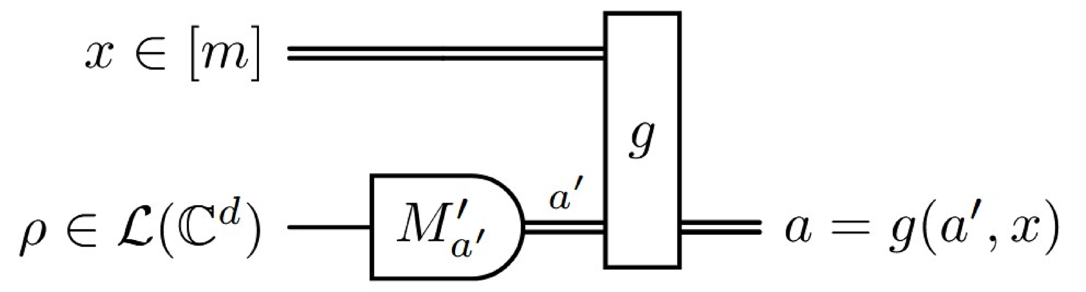
Measurement Incompatibility



Objective: Simulate measurement assemblage $\mathcal{M} := \{\mathrm{M}_{a|x}\}_{a,x}$ using one parent POVM $\{\mathrm{M}'_{a'}\}_{a'}$ and classical post-processing.

$\Longrightarrow$ **Reduction of m measurements to 1 measurement.**

Definition:

$$\mathrm{M}_{a|x} = \sum_{a'} q(a|x,a')\mathrm{M}'_{a'} \iff \sum_{\vec{a}\setminus a_x} \mathrm{M}'_{\vec{a}} = \mathrm{M}_{a|x}.$$

- Post-processing $q(a|x,a')$ can be reduced to deterministic functions $a = g(a',x)$.
- $\{\mathrm{M}'_{\vec{a}}\}_{\vec{a}}$ is a parent POVM with outcomes $[a_{x=1}, a_{x=2}, \cdots, a_{x=m}]$.

Example:

$$\mathrm{M}^{\eta}_{\pm|1} = \frac{1}{2}(\mathbb{1} \pm \eta\sigma_x),\ \mathrm{M}^{\eta}_{\pm|2} = \frac{1}{2}(\mathbb{1} \pm \eta\sigma_z).$$

Noise Model:

$$\mathrm{M}^{\eta}_{a|x} = \eta \mathrm{M}_{a|x} + (1-\eta)\mathrm{Tr}[\mathrm{M}_{a|x}]\frac{\mathbb{1}}{d}.$$

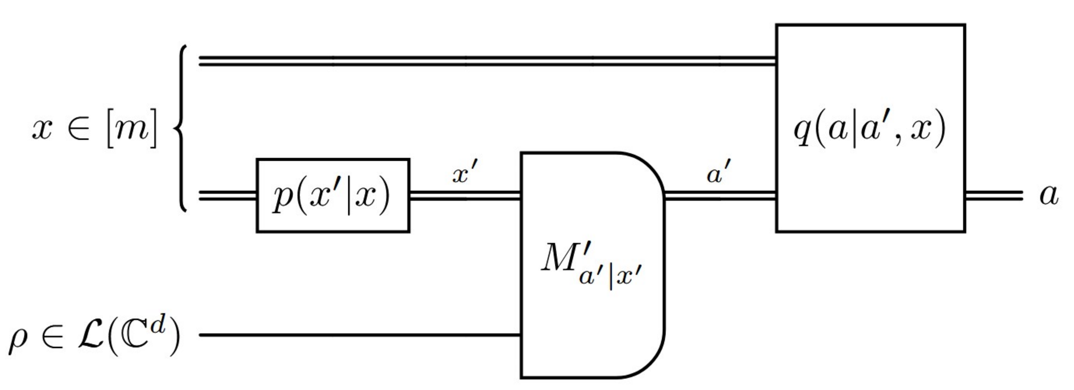
- Find maximal visibility parameter $\eta \in [0,1]$ such that $\mathcal{M}_1, \mathcal{M}_2$ are jointly measurable.
- Candidate parent POVM: $\mathrm{M}'^{\eta}_{i,j} = \frac{1}{4}(\mathbb{1} + \eta(i\sigma_x + j\sigma_z))$ where $i, j = \pm 1$.
- Leads to maximal $\eta^* = \frac{1}{\sqrt{2}}$, which can be shown to be optimal.

Simulability and Compatibility Structures

Main Questions:

1) What does it mean for an assemblage of m settings to contain *genuinely* n measurements?
2) How do the different approaches to 1) *compare* to each other?

n-wise measurement simulability:



Definition [1]:

$$\mathrm{M}_{a|x} = \sum_{x'=1}^{n} p(x'|x) \sum_{a'} \mathrm{M}'_{a'|x'} q(a|x,a').$$

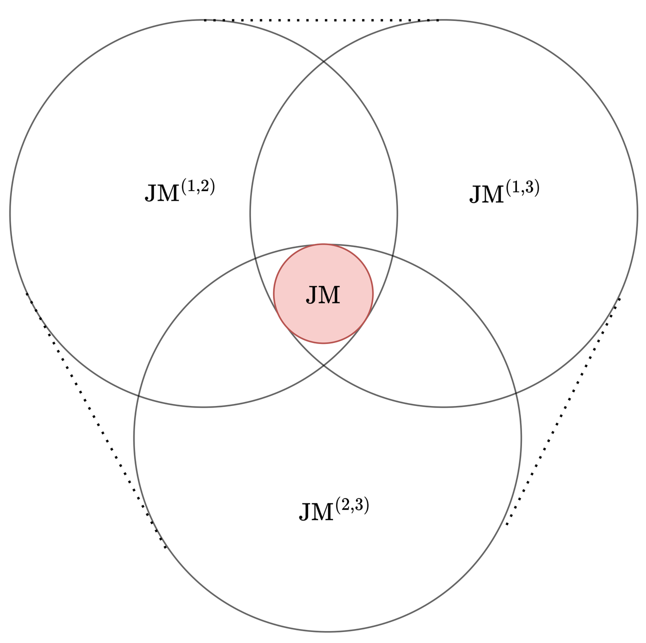
Extension:

$$\mathrm{M}_{a|x} = \sum_{\lambda} \pi(\lambda) \sum_{x'=1}^{n} p(x'|x,\lambda) \sum_{a'} \mathrm{M}'_{a'|x'} q(a|x,a',\lambda).$$

- Operationally well-defined $\Longrightarrow$ **Classical pre-and post-processing** $+$ **available measurements**.
- Hard to analyze due to non-linearity and unclear geometric structure.

n-wise compatible structures [2]:

- Full joint measurability JM
- One JM pair: $\mathrm{JM}^{(1,2)}, \mathrm{JM}^{(2,3)}, \mathrm{JM}^{(1,3)}$.
- Pairwise JM: $\mathrm{JM}^{(1,2)} \cap \mathrm{JM}^{(2,3)} \cap \mathrm{JM}^{(1,3)}$
- 2-wise compatible: $\mathrm{Conv}(\mathrm{JM}^{(1,2)}, \mathrm{JM}^{(1,3)}, \mathrm{JM}^{(2,3)})$



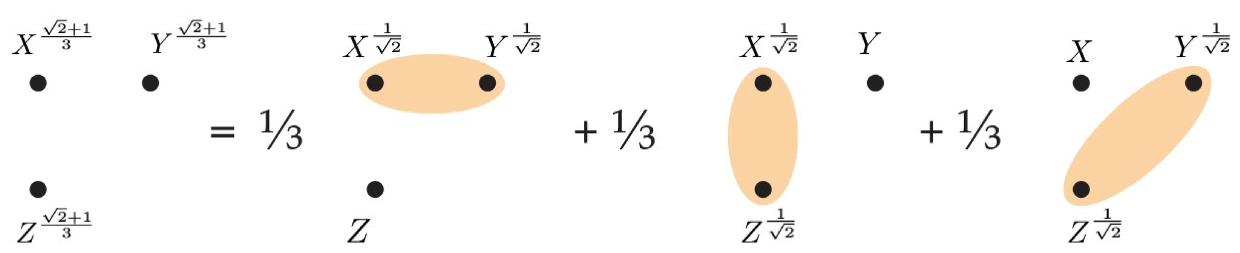
Definition of (m-1) compatible:

$$\mathrm{M}_{a|x} = \sum_{(t_1,t_2\neq t_1)\in\mathcal{T}} p_{(t_1,t_2)} \mathrm{J}^{(t_1,t_2)}_{a|x}.$$

Example:

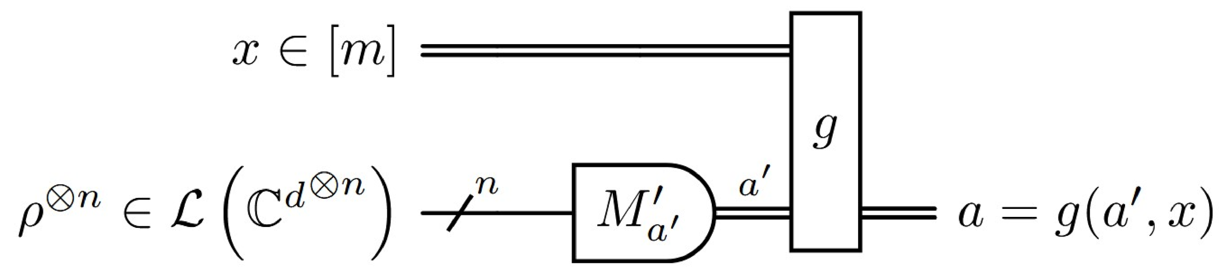
$$\mathrm{M}^{\eta}_{\pm|s} = \frac{1}{2}(\mathbb{1} \pm \eta\sigma_s) \text{ with } s \in \{x,y,z\}.$$

- JM: $\eta \leq 1/\sqrt{3} \approx 0.58$.
- Pairwise JM: $\eta \leq 1/\sqrt{2} \approx 0.71$.
- 2-wise compatible: $\eta \leq \frac{\sqrt{2}+1}{3} \approx 0.80$.



$\Longrightarrow$ Questions the role of (free) randomness and its operational meaning.

Multi-Copy Incompatibility



Definition [3]:

$$\mathrm{Tr}[\mathrm{M}_{a|x}\rho] = \sum_{a'} p(a|x,a')\mathrm{Tr}[\mathrm{M}'_{a'}\rho^{\otimes n}]\ \forall \rho.$$

Example:

$$\mathrm{M}^{\eta}_{\pm|s} = \frac{1}{2}(\mathbb{1} \pm \eta\sigma_s) \text{ with } s \in \{x,y,z\}.$$
$$\Longrightarrow \eta \leq \sqrt{3}/2.$$

Main Results

A strict hierarchy:

$$\mathrm{JM} \subset \mathrm{SIM}^{\mathrm{Det}}_n \subset \mathrm{SIM}_n \subset \mathrm{Conv}(\mathrm{SIM}_n) \equiv \mathrm{JM}^{\mathrm{conv}}_n \subset \mathrm{Copy}_n \subset \mathrm{All}_m.$$

- $\mathrm{JM} \equiv$ jointly measurable.
- $\mathrm{SIM}^{\mathrm{Det}}_n \equiv$ deterministic n-simulability.
- $\mathrm{SIM}_n \equiv n$-simulability.
- $\mathrm{Conv}(\mathrm{SIM}_n) \equiv$ Convex hull of SIM_n.
- $\mathrm{JM}^{\mathrm{conv}}_n \equiv n$-wise compatible measurements.
- $\mathrm{Copy}_n \equiv n$-copy joint measurability.

$\Longrightarrow$ **Different notions of what *genuinely* constitutes n measurements can be ordered into a hierarchy.**
$\Longrightarrow$ **Operational vs. geometrical interpretation, the choice of the right notion depends on application.**
$\Longrightarrow$ **Correspondence between different models via: $\mathrm{Conv}(\mathrm{SIM}_n) \equiv \mathrm{JM}^{\mathrm{conv}}_n$.**

$\mathrm{SIM}^{\mathrm{Det}}_n \subset \mathrm{SIM}_n$

- $\mathrm{SIM}^{\mathrm{Det}}_n$ can be described by SDP constraints.
- Consider σ_x, σ_z and Hadamard measurements.
- Find good probabilistic simulation.
- $\eta_{\mathrm{det}} \approx 0.7654$ and $\eta_{\mathrm{prob}} \approx 0.8150$.

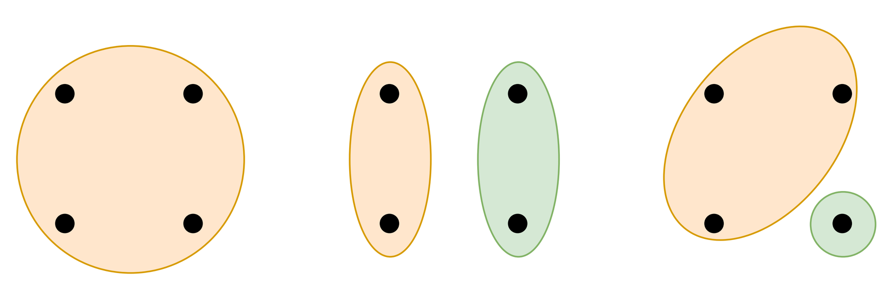
$\mathrm{SIM}_n \subset \mathrm{Conv}(\mathrm{SIM}_n)$

- Consider $\sigma_x, \sigma_y, \sigma_z$ measurements.
- Find explicit construction for $\mathrm{Conv}(\mathrm{SIM}_n)$.
- Approximate SIM_n through ϵ-grid for pre-processing
- Find error estimate ν s.t. $\eta_{\mathrm{grid}} + \nu < \eta_{\mathrm{Conv}(\mathrm{SIM}_n)}$.

$\mathrm{Conv}(\mathrm{SIM}_n) \equiv \mathrm{JM}^{\mathrm{conv}}_n$

$$\mathrm{Conv}(\mathrm{SIM}_n): \ \mathrm{M}_{a|x} = \sum_{\lambda} \pi(\lambda) \sum_{x'=1}^{n} p^{\lambda}(x'|x) \sum_{a'} q^{\lambda}(a|a',x)\mathrm{M}'^{\lambda}_{a'|x'}.$$

- Consider first deterministic pre-processing $p^{\lambda}(x'|x)$.
- Correspondence to compatibility hypergraphs $\Longrightarrow$ Equivalence follows by design.
- Probabilistic pre-processing $p^{\lambda}(x'|x)$ can be made deterministic by re-defining $\pi(\lambda)$.



$\mathrm{JM}^{\mathrm{conv}}_n \subset \mathrm{Copy}_n$

- Use known [4] relation $\mathrm{SIM}_n \implies \mathrm{Copy}_n$.
- Copy_n is a convex set that contains SIM_n. Hence, $\mathrm{JM}^{\mathrm{conv}}_n \subseteq \mathrm{Copy}_n$.
- Strict separation follows from considering noisy $\sigma_x, \sigma_y, \sigma_z$ measurements.

A universal lower bound on multi-copy incompatibility:

Given any assemblage $\mathcal{M}$ of m measurements of dimension d, it holds for the critical n-copy visibility:

$$\eta_{\mathrm{Copy}_n} \geq \frac{n(d+m)}{m(d+n)}.$$

- Use approximate broadcasting [5] from $n \to m$ copies.
- m measurements are trivially JM on m copies.
- Noiseless measurements on noisy states $\iff$ Noisy measurements on noiseless states.

Implications and Conclusions

- Different notions exist, answer depends on the application in mind.
- We fully resolved previously unnoticed ambiguity in the literature.
- Direct implications for the (semi)-device-independent certification of number of measurements [6, 7].

Open questions:

- Do other operationally meaningful generalizations of joint measurability exist?
- How to certify n-copy incompatibility device-independently?

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